MATH 345 Skeleton Notes
Unit 4: Orthogonality and least squares
Section 4.4: QR factorization and least squares

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From Gram–Schmidt to QR

The Gram-Schmidt algorithm processes the independent columns of a matrix from left to right. It constructs orthonormal vectors $\mathbf{q}_1, \dots, \mathbf{q}_n$ and records coefficients r_{ij}. These same quantities form a matrix factorization.

Let

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}.$$

At column j, the Gram–Schmidt algorithm computes

$$\mathbf{p}_j = \sum_{i<j} r_{ij}\mathbf{q}_i, \qquad \mathbf{v}_j = \mathbf{a}_j - \mathbf{p}_j,$$

$$r_{jj} = \|\mathbf{v}_j\|, \qquad \mathbf{q}_j = \frac{\mathbf{v}_j}{r_{jj}}.$$

Since

$$\mathbf{v}_j = r_{jj}\mathbf{q}_j,$$

we can reconstruct the original column:

$$\mathbf{a}_j = \mathbf{p}_j + \mathbf{v}_j = \sum_{i<j} r_{ij}\mathbf{q}_i + r_{jj}\mathbf{q}_j = \sum_{i\le j} r_{ij}\mathbf{q}_i.$$

Thus column j uses only $\mathbf{q}_1, \dots, \mathbf{q}_j$.
Collect the orthonormal vectors in

$$Q = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \end{bmatrix}$$

and the coefficients in

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ 0 & r_{22} & \cdots & r_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & r_{nn} \end{bmatrix}.$$

The column equations above become the single matrix equation

$$A = QR.$$

The zeros below the diagonal of R record the order of the algorithm: column j uses only directions constructed by stage j.

Definition: Reduced QR factorization

Let A be an $m \times n$ matrix with

linearly independent columns.

A reduced QR factorization of A is a factorization

$$A = QR,$$

where

- Q is an $m \times n$ matrix with orthonormal columns;
- R is an invertible upper-triangular $n \times n$ matrix.

For the Gram–Schmidt convention used in this unit, the diagonal entries of R are positive:

$$r_{jj} = \|\mathbf{v}_j\| > 0.$$

Activity: Reading Q^T Q (U4-LO1)

Let

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{6} \\ 1/\sqrt{3} & -2/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{6} \end{bmatrix} = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 \end{bmatrix}.$$

1. Compute $\|\mathbf{q}_1\|^2$ and $\|\mathbf{q}_2\|^2$.
2. Compute $\mathbf{q}_1^T \mathbf{q}_2$.
3. Compute $Q^T Q$.
4. What does the (i, j)-entry of $Q^T Q$ measure?
5. Explain why

$$Q^T Q = I_2$$

is the matrix check that the columns of Q are orthonormal.

We have

$$\|\mathbf{q}_1\|^2 = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1$$

and

$$\|\mathbf{q}_2\|^2 = \frac{1}{6} + \frac{4}{6} + \frac{1}{6} = 1.$$

Also,

$$\mathbf{q}_1^T \mathbf{q}_2 = \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} + \frac{1}{\sqrt{18}} = 0.$$

Therefore

$$Q^T Q = \begin{bmatrix} \mathbf{q}_1^T \mathbf{q}_1 & \mathbf{q}_1^T \mathbf{q}_2 \\ \mathbf{q}_2^T \mathbf{q}_1 & \mathbf{q}_2^T \mathbf{q}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

The (i, j)-entry of $Q^T Q$ is the dot product $\mathbf{q}_i^T \mathbf{q}_j$. The diagonal entries check unit length, and the off-diagonal entries check orthogonality.

Theorem: Gram–Schmidt produces QR

Every $m \times n$ matrix A with

linearly independent columns

has a reduced QR factorization

$$A = QR$$

in which the columns of Q are orthonormal and R is upper triangular with positive diagonal entries.

Why is this true?

Apply the Gram-Schmidt algorithm to the columns

$$\mathbf{a}_1, \dots, \mathbf{a}_n.$$

At stage j, the algorithm gives

$$\mathbf{a}_j = \sum_{i \le j} r_{ij} \mathbf{q}_i.$$

Collecting these equations column by column gives

$$A = QR.$$

The columns of Q are orthonormal. The condition $i \le j$ makes R upper triangular, and

$$r_{jj} = \|\mathbf{v}_j\| > 0$$

because the input columns are linearly independent.

Corollary: Square case

Every invertible square matrix has a QR factorization

$$A = QR,$$

where Q is an orthogonal matrix and R is upper triangular with positive diagonal entries.

Why is this true?

An invertible square matrix has linearly independent columns. Apply "Gram–Schmidt produces QR." Since Q is square and has orthonormal columns,

$$Q^T Q = Q Q^T = I.$$

Thus Q is an orthogonal matrix.

Activity: Build Q and R from the Gram–Schmidt ledger (U4-LO2, U4-LO6)

Let

$$A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix}.$$

1. Compute

$$r_{11} = \|\mathbf{a}_1\|$$

and

$$\mathbf{q}_1 = \frac{\mathbf{a}_1}{r_{11}}.$$

2. Compute the projection coefficient

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2.$$

3. Compute

$$\mathbf{p}_2 = r_{12}\mathbf{q}_1$$

and

$$\mathbf{v}_2 = \mathbf{a}_2 - \mathbf{p}_2.$$

4. Compute

$$r_{22} = \|\mathbf{v}_2\|$$

and

Activity: Build Q and R from the Gram–Schmidt ledger (U4-LO2, U4-LO6) (continued)

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{r_{22}}.$$

5. Assemble

$$Q = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 \end{bmatrix}$$

and

$$R = \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix}.$$

6. Check

$$Q^T Q = I_2$$

and

$$QR = A.$$

The first column gives

$$r_{11} = \|\mathbf{a}_1\| = 5, \qquad \mathbf{q}_1 = \frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

The projection coefficient is

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2 = \frac{3}{5}(1) + \frac{4}{5}(2) = \frac{11}{5}.$$

Therefore

$$\mathbf{p}_2 = r_{12}\mathbf{q}_1 = \frac{11}{5}\begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 33/25 \\ 44/25 \end{bmatrix},$$

and

$$\mathbf{v}_2 = \mathbf{a}_2 - \mathbf{p}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 33/25 \\ 44/25 \end{bmatrix} = \begin{bmatrix} -8/25 \\ 6/25 \end{bmatrix}.$$

Its length is

$$r_{22} = \|\mathbf{v}_2\| = \frac{2}{5},$$

so

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{r_{22}} = \begin{bmatrix} -4/5 \\ 3/5 \end{bmatrix}.$$

Thus

$$Q = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}, \qquad R = \begin{bmatrix} 5 & 11/5 \\ 0 & 2/5 \end{bmatrix}.$$

The orthonormality check gives

$$Q^T Q = I_2,$$

and direct multiplication gives

$$QR = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix} = A.$$

Activity: An upper-triangular matrix already has a QR factorization (U4-LO6)

Let

$$A = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}.$$

Explain why

$$Q = I_2, \qquad R = A$$

is a QR factorization. Check each condition in the definition.

The columns of I_2 are orthonormal, so

$$Q^T Q = I_2.$$

The matrix

$$R = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}$$

is upper triangular and has positive diagonal entries. Finally,

$$QR = I_2 A = A.$$

Thus $A = QR$ is a QR factorization.

Warning: Rectangular Q

In a reduced QR factorization of an $m \times n$ matrix with $m > n$, the matrix Q has orthonormal columns, so

$$Q^T Q = I_n.$$

However, Q is not square, and generally

$$QQ^T \neq I_m.$$

Instead,

$$QQ^T \mathbf{b} = \text{proj}_{\text{col}(Q)}(\mathbf{b}) = \text{proj}_{\text{col}(A)}(\mathbf{b}).$$

Indeed, the entries of $Q^T \mathbf{b}$ are the dot products $\mathbf{q}_i \cdot \mathbf{b}$. Multiplying by Q forms the corresponding linear combination of its columns:

$$QQ^T\mathbf{b} = \sum_{i=1}^{n}(\mathbf{q}_i \cdot \mathbf{b})\mathbf{q}_i = \text{proj}_{\text{col}(Q)}(\mathbf{b}).$$

The last equality follows from the Projection theorem, since the columns of Q form an orthonormal basis of $\text{col}(Q)$.
Also, $A = QR$ expresses every column of A as a linear combination of the columns of Q. Since R is invertible, $Q = AR^{-1}$ gives the reverse inclusion. Thus $\text{col}(Q) = \text{col}(A)$, so the two projections agree.

Warning: Signs in a QR factorization

The positive-diagonal convention fixes the signs of the Gram–Schmidt factorization. Without that convention, one may replace a column $\mathbf{q}_j$ by $-\mathbf{q}_j$ and replace row j of R by its negative. The product QR does not change. Numerical software may therefore return Q and R with signs different from a hand calculation. Check $Q^TQ = I$ and $QR = A$ rather than requiring every entry to have the same sign.

Least squares with QR

The normal equations in Least squares and the normal equations give a direct route from a QR factorization to a least-squares solution.
Suppose

$$A = QR$$

is a reduced QR factorization. The normal equations are

$$A^TA\hat{\mathbf{x}} = A^T\mathbf{b}.$$

Substituting $A = QR$ gives

$$R^TQ^TQR\hat{\mathbf{x}} = R^TQ^T\mathbf{b}.$$

Since

$$Q^TQ = I,$$

this becomes

$$R^TR\hat{\mathbf{x}} = R^TQ^T\mathbf{b}.$$

The matrix R is invertible, so R^T is invertible. Canceling R^T gives

$$R\hat{\mathbf{x}} = Q^T\mathbf{b}.$$

The next activity shows why this upper-triangular system is easy to solve by back-substitution, the method introduced in Solving a system using back substitution (U2-LO1, U2-LO2) in Unit 2.

Activity: Back-substitution (U2-LO2)

Solve

$$R\mathbf{x} = \mathbf{y},$$

where

$$R = \begin{bmatrix} 4 & 2 & 7 \\ 0 & 3 & 5 \\ 0 & 0 & 2 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 29 \\ 21 \\ 6 \end{bmatrix}.$$

Work from the last equation upward.

The last equation gives

$$2x_3 = 6, \qquad x_3 = 3.$$

The second equation gives

$$3x_2 + 5x_3 = 21,$$

so

$$3x_2 + 15 = 21, \qquad x_2 = 2.$$

The first equation gives

$$4x_1 + 2x_2 + 7x_3 = 29,$$

so

$$4x_1 + 4 + 21 = 29, \qquad x_1 = 1.$$

Therefore

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$$

Algorithm: Least squares with QR

Input. An $m \times n$ matrix A with linearly independent columns and a vector $\mathbf{b} \in \mathbb{R}^m$.

1. Compute a reduced QR factorization

$$A = QR.$$

2. Compute

$$\mathbf{y} = Q^T \mathbf{b}.$$

3. Solve the upper-triangular system

$$R\hat{\mathbf{x}} = \mathbf{y}$$

by back-substitution.

Output. The vector $\hat{\mathbf{x}}$ is the least-squares coefficient vector. The fitted vector is

$$A\hat{\mathbf{x}} = QR\hat{\mathbf{x}} = QQ^T \mathbf{b}.$$

This method avoids forming $A^T A$ or computing a matrix inverse: once Q and R are known, we need only dot products and back-substitution. If A stays fixed while $\mathbf{b}$ changes, we can reuse the same factorization for each new right-hand side. In floating-point computation, a numerically stable QR routine can also be more accurate than solving the normal equations, especially when the columns of A are nearly linearly dependent.

Activity: Solving least squares with QR (U4-LO2, U4-LO6)

Use QR to fit a line $y = c_0 + c_1 t$ to the data points

$$(0, 1), \qquad (1, 2), \qquad (2, 2)$$

from Fit a line to three data points (U4-LO4, U4-LO5). The design matrix and data vector are

Activity: Solving least squares with QR (U4-LO2, U4-LO6) (continued)

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

1. Apply Gram–Schmidt to the columns of X to find a reduced factorization $X = QR$ with positive diagonal entries in R.
2. Compute $Q^T\mathbf{y}$.
3. Solve $R\hat{\mathbf{c}} = Q^T\mathbf{y}$ by back-substitution and write the fitted line.

Write $X = [\mathbf{a}_1\ \mathbf{a}_2]$. The first column gives

$$r_{11} = \|\mathbf{a}_1\| = \sqrt{3}, \qquad \mathbf{q}_1 = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

For the second column,

$$r_{12} = \mathbf{q}_1^T\mathbf{a}_2 = \frac{0+1+2}{\sqrt{3}} = \sqrt{3}.$$

Subtracting its component along $\mathbf{q}_1$ gives

$$\mathbf{v}_2 = \mathbf{a}_2 - r_{12}\mathbf{q}_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Hence

$$r_{22} = \|\mathbf{v}_2\| = \sqrt{2}, \qquad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

Thus the reduced factors are

$$Q = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}, \qquad R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{2} \end{bmatrix}.$$

The transformed right-hand side is

$$Q^T \mathbf{y} = \begin{bmatrix} \mathbf{q}_1^T \mathbf{y} \\ \mathbf{q}_2^T \mathbf{y} \end{bmatrix} = \begin{bmatrix} (1+2+2)/\sqrt{3} \\ (-1+0+2)/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 5/\sqrt{3} \\ 1/\sqrt{2} \end{bmatrix}.$$

The last equation of $R\hat{\mathbf{c}} = Q^T \mathbf{y}$ gives

$$\sqrt{2}\,\hat{c}_1 = \frac{1}{\sqrt{2}}, \qquad \hat{c}_1 = \frac{1}{2}.$$

Substituting into the first equation gives

$$\sqrt{3}\,\hat{c}_0 + \sqrt{3}\left(\frac{1}{2}\right) = \frac{5}{\sqrt{3}},$$

so

$$\hat{c}_0 = \frac{5}{3} - \frac{1}{2} = \frac{7}{6}.$$

Therefore

$$\hat{\mathbf{c}} = \begin{bmatrix} 7/6 \\ 1/2 \end{bmatrix},$$

and the fitted line is

$$y = \frac{7}{6} + \frac{1}{2}t.$$

Activity: Solving least squares with QR by code (U4-LO6, U4-LO7)

The following code repeats the *QR* least-squares calculation from Solving least squares with *QR* (U4-LO2, U4-LO6).
The function `np.linalg.qr` returns the two factors, which the assignment stores in `Q` and `R`: `Q` has orthonormal columns, `R` is upper triangular, and their product is `X`. The option `mode="reduced"` keeps only the columns of `Q` needed to span the column space of `X`, together with the corresponding square factor `R`, rather than extending `Q` to a square orthogonal matrix.
The expression `np.eye(Q.shape[1])` constructs an identity matrix whose size is the number of columns of `Q`; here it is the 2×2 identity matrix.
The function `np.linalg.solve` solves a square system directly,

Activity: Solving least squares with QR by code (U4-LO6, U4-LO7) (continued)

without explicitly computing the inverse of its coefficient matrix. Using it is preferable to forming an inverse and then multiplying, which adds unnecessary computation and can introduce additional rounding error.

```
import numpy as np

t = np.array([0.0, 1.0, 2.0])
y = np.array([1.0, 2.0, 2.0])

X = np.column_stack([np.ones_like(t), t])

Q, R = np.linalg.qr(X, mode="reduced")

c_qr = np.linalg.solve(R, Q.T @ y)
c_lstsq = np.linalg.lstsq(X, y, rcond=None)[0]

(
    Q.shape,
    R.shape,
    np.allclose(Q.T @ Q, np.eye(Q.shape[1])),
    np.allclose(Q @ R, X),
    c_qr,
    c_lstsq,
    np.allclose(c_qr, c_lstsq),
)
```

1. What should `Q.shape` and `R.shape` be?
2. What does Q.T @ Q
   check?
3. What does Q @ R
   check?

Activity: Solving least squares with QR by code (U4-LO6, U4-LO7) (continued)

4. What mathematical vector is computed by Q.T @ y ?
5. Why does the code solve a system with R?
6. Why should `c_qr` and `c_lstsq` agree?
7. Does their agreement mean that

$$X\mathbf{c} = \mathbf{y}$$

   has an exact solution?
8. Why may the entries of Q and R have signs different from a hand Gram–Schmidt calculation?

Since X has shape `(3, 2)`, reduced QR gives

```
Q.shape == (3, 2)
R.shape == (2, 2)
```

The expression

```
Q.T @ Q
```

checks whether the columns of Q are orthonormal. The expression

```
Q @ R
```

checks whether the factors reconstruct `X`.
The vector

```
Q.T @ y
```

is the right-hand side $Q^T \mathbf{y}$ in the reduced system

$$R\hat{\mathbf{c}} = Q^T \mathbf{y}.$$

The matrix `R` is square, invertible, and upper triangular, so the code solves this system with `np.linalg.solve`.
Both methods compute the same least-squares coefficient vector, so

```
c_qr
```

and

```
c_lstsq
```

should both be close to

```
array([1.16666667, 0.5])
```

and the final `np.allclose` check should return `True`.
This agreement does not mean that

$$X\mathbf{c} = \mathbf{y}$$

has an exact solution. The residual from the Fit a line to three data points (U4-LO4, U4-LO5) is nonzero.
A valid QR factorization can use simultaneous sign changes in one column of Q and the corresponding row of R. Therefore numerical output may have different signs from a hand calculation while still satisfying

$$Q^T Q = I \qquad \text{and} \qquad QR = X.$$